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Phenology aware agricultural boundary extraction using segment anything model and planet scope imagery (zero shot learning approach)

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Abstract

Accurate identification of agricultural parcel boundaries is critical for sustainable agricultural management, land use planning, and digital agriculture applications. Especially in advanced analyses such as precision agriculture, crop monitoring and yield estimation, reliable parcel data constitute an essential analytical component. The Segment Anything Model (SAM), a recent advancement in visual semantic segmentation, demonstrates significant potential for the automatic extraction of agricultural parcels from high-resolution satellite imagery. However, several challenges remain regarding the model's applicability across diverse agricultural contexts, including its parameterization, pre-processing requirements, and integration into operational workflows. The primary objective of this study is to integrate phenology-driven multi-temporal image selection with the zero-shot segmentation capability of SAM for the automatic delineation of agricultural parcel boundaries. Within the scope of the study, NDVI time series obtained from Sentinel-2 satellite data were overlaid with farmer-declared parcels to create characteristic spectral profiles for each crop. From these profiles, three dates (May 25, July 19 and September 22, 2022) that best represent the phenological stages of the crops were selected and PlanetScope images of these dates were used. Combinations of true color, false color and NDVI time series (NDVI TS) were generated from the images and provided as input to the SAM model. The results show that the NDVI TS approach outperforms the single date segmentation. The multi-temporal segmentation achieved an IoU of 0.89 and F1 score of 0.93, while the geometric segmentation errors remained low (GOSE: 0.12; GUSE: 0.13). In single date analyses, IoU values remained in the range of 0.73–0.81 and errors increased, especially during periods of poor vegetation cover. This study shows that phenologically-based multi-temporal image selection significantly improves segmentation performance and that zero-shot models such as SAM can perform highly accurate and operationally feasible boundary extraction without the need for large scale training data. Supported by high-resolution satellite data, this approach provides an efficient, cost-effective and directly applicable method for identifying cultivated land patterns and types, emphasizing that besides the model, the content characteristics and acquisition time of the imagery are also determining factors in segmentation success.

Keywords: Boundary; Crop; Phenology; PlanetScope; SAM

1. Introduction

Precise delineation of agricultural land boundaries is critical for ensuring sustainable agricultural management

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at local and global scales, supporting food security policies, optimizing land use planning, and increasing agricultural production efficiency. Agricultural parcel boundaries serve as the fundamental spatial unit in a wide range of decision-support applications, from production statistics to environmental impact assessments. Well-defined field boundaries not only contribute to the agricultural economy but also serve as key parameters in water resource management, soil erosion modeling, rural socio-economic analysis, and post-disaster recovery efforts (Campbell, 2011; Estes et al., 2016; Garcia-Pedrero et al., 2018). In recent years, remote sensing technologies have played an increasingly pivotal role in agricultural monitoring. Medium-resolution imagery (e.g., Sentinel-2) facilitates systematic agricultural observation at regional to national scales, while high-resolution imagery (e.g., PlanetScope, WorldView, Gaofen) offers critical insights for fine-scale and parcel-level analyses (North et al., 2018; Weiss et al., 2020). These data sources have become central to applications such as crop type classification, harvest timing estimation, vegetation health monitoring, land use change detection, and yield estimation. However, the automatic extraction of agricultural parcel boundaries from such imagery constitutes a complex spatial modeling task rather than a straightforward classification problem. It involves object-based analytical methods, multi-dimensional data processing, and the interpretation of contextual visual cues (Crommelinck et al., 2016; Watkins and Van Niekerk, 2019). Accurately delineating boundaries requires the integration of spectral, textural, edge-based, and contextual information. While pixel-based classifications often overlook the spatial coherence of parcels, object-based approaches yield outputs that are more aligned with decision-support systems. Consequently, field-scale segmentation is generally favored over pixel-based methods in agricultural mapping workflows (North et al., 2018). Traditionally, manual digitization and rule-based segmentation algorithms (e.g., edge-based, region-based, or hybrid approaches) have been employed to define agricultural boundaries. However, edge-based methods are prone to over-segmentation, and region-based techniques often suffer from high sensitivity to parameter settings, resulting in suboptimal performance in heterogeneous or small-scale landscapes (Zhang et al., 2021; Cai et al., 2022).

These limitations become more pronounced in regions with fragmented and smallholder-dominated agriculture. With the advancement of spatial resolution in satellite imagery and increased access to geospatial data, machine learning and classical computer vision techniques have become prominent tools for parcel boundary detection. Various segmentation paradigmsedge-based, region-based, and hybrid have been widely applied (Yan and Roy, 2014; Garcia-Pedrero et al., 2017; Graesser and Ramankutty, 2017; Crommelinck et al., 2017; Cai et al., 2022; Duvvuri and Kambhammettu, 2023). Nevertheless, each method has intrinsic drawbacks: edge-based approaches may lead to over-segmentation in high-resolution images, while region-based methods may yield poor results if parameter tuning is not well adapted to the landscape (Watkins and Van Niekerk, 2019; Zhang et al., 2021). To overcome these challenges, deep learning-based segmentation models have emerged as state-of-the-art solutions in remote sensing image analysis. Architectures such as U-Net, DeepLab, and Mask R-CNN have demonstrated strong performance in field boundary extraction tasks (Persello et al., 2019; Waldner and Diakogiannis, 2020; Long et al., 2015). These models are capable of capturing contextual relationships, recognizing complex visual patterns, and producing high-resolution semantic segmentation maps. However, their widespread deployment is hindered by several limitations: they typically require extensive labeled datasets, demand significant computational resources, and often necessitate task-specific retraining or fine-tuning (Tripathy et al., 2024). These constraints limit the accessibility of such models, particularly in resource-constrained regions and countries characterized by small-scale, diversified farming systems. For instance, agricultural zones comprising parcels smaller than 0.6 ha, such as in Bangladesh, Nigeria, and the Gangetic Plains of India, require specialized solutions in terms of spatial resolution and model sensitivity (Zhang et al., 2020). As a response to these limitations, foundation models have gained substantial attention. Pre-trained on large and diverse datasets, these models can be adapted to new tasks through zero-shot or few-shot learning strategies without extensive retraining.

The Segment Anything Model (SAM), developed by Meta AI and released in 2023, stands as a notable example of a vision foundation model. Based on a Vision Transformer (ViT) architecture, SAM was trained on approximately 11 million images and over 1 billion segmentation masks. It enables high-accuracy segmentation with simple user prompts (e.g., point, box, or text) and does not require task-specific fine-tuning (Kirillov et al., 2023). Among its key advantages are independence from labeled data, fast execution, user-guided flexibility, and adaptability to images of varying spatial resolutions. These features have positioned SAM as a promising tool for automated delineation of agricultural field boundaries. Recent studies support its efficacy: Tripathy et al. (2024) demonstrated that SAM can accurately map smallholder parcels using a zero-shot approach, outperforming traditional deep learning models with minimal data input. Huang et al. (2025) showed that multi-scale segmentation strategies integrated with SAM improve boundary precision in high-resolution imagery. Huang et al. (2024) further illustrated that SAM can be used in conjunction with time-series data to monitor land use changes. Ferreira et al. (2025) introduced the FieldSeg framework, which combined SAM with Sentinel-2 imagery and achieved up to 48.6 % object-based overall accuracy (OBOA) and an average Dice Similarity Coefficient (DSC) of 0.749 across eight agro-ecological zones. Particularly for large parcels (≥5 ha), boundary accuracy reached 81.4 %. Similarly, Sun et al. (2025) proposed the SIDEST method, which integrates a double-edge-corrected SAM approach with super-resolution Sentinel-2 imagery. The model achieved an IoU of 0.84 using upsampled (from 10 m to 3 m) images, even without reference data. Despite these advancements, the influence of SAM's internal configurations (e.g., ViT-B, ViT-L, ViT-H variants), input image resolution, seasonal imagery selection, and pre-segmentation enhancements on model performance remains underexplored. Hence, comprehensive comparative analyses using diverse satellite platforms (e.g., Sentinel-2, PlanetScope, Gaofen) and various spectral inputs (e.g., vegetation indices such as NDVI, EVI) are necessary. Such studies will contribute to a more systematic understanding of SAM's applicability across different geographies and agricultural systems (Ferreira et al., 2025; Liu, 2024).

This study aims to evaluate the performance of the Segment Anything Model (SAM) in delineating agricultural parcel boundaries through a comprehensive and multi-faceted approach. Using high-resolution PlanetScope (PSB.SD) satellite imagery, both single-date and multi-temporal (time-series) datasets were analyzed. Several image-enhancement techniques were applied to the model inputs to improve visual distinctiveness and segmentation accuracy. The study particularly investigates the zero-shot learning capability of SAM and assesses its effectiveness in detecting field boundaries within small-scale, heterogeneous, and phenologically diverse agricultural landscapes.

This research demonstrates that, in the automatic delineation of agricultural field boundaries using remote-sensing data, not only the performance of the model but also the acquisition timing and spectral characteristics of the satellite imagery play a decisive role. Understanding the crop pattern within the study area and employing images captured during different phenological stages of dominant crops enhance spectral separability among vegetation types, resulting in clearer and more distinguishable field boundaries. Composite images derived from time-series data provide a more comprehensive representation of surface spectral characteristics, while temporal information accurately reflects crop-growth dynamics, thereby strengthening boundary-delineation performance.

All datasets were obtained from PlanetScope (PSB.SD) imagery acquired on three phenologically representative dates with high temporal relevance. Prior to segmentation, advanced image-enhancement techniques—such as histogram equalization, contrast stretching, and edge sharpening—were applied to increase spectral contrast and improve class separability. Importantly, the temporal image-selection process was not arbitrary; instead, the periods with maximum spectral separability were scientifically identified based on the phenological stages of the dominant crops in the study area. This approach enabled a systematic evaluation of SAM's sensitivity to phenological dynamics under zero-shot conditions, encompassing both temporal and spectral dimensions at the parcel level. Unlike previous studies that primarily relied on single-date imagery with minimal preprocessing, this research integrates enriched, index-based, and multi-temporal datasets with advanced enhancement techniques. The findings indicate that SAM can achieve high segmentation accuracy, particularly in agricultural production areas characterized by pronounced phenological variability. Ultimately, this study reveals that phenology-aware, multi-temporal, and spectrally enriched datasets can optimize the zero-shot capability of SAM, offering an innovative and operational framework for automated agricultural boundary delineation.

2. Study area and materials

2.1. Study area

The study area is located in the Hotamış region of the Karapınar district, within Konya province in the Central Anatolia Region of Turkey. Geographically, it lies approximately between 37°30′N–37°50′N latitude and 33°15′E–33°27′E longitude, covering a total area of around 250 km2 (Fig. 1). The region falls within the Konya Closed Basin and is characterized by semi-arid climatic conditions. The average annual temperature ranges between 11 °C and 12 °C, while annual precipitation amounts to approximately 300–350 mm. Rainfall is concentrated in the spring months, whereas the summer season is typically hot and dry. The Hotamış region is predominantly composed of expansive and flat agricultural lands that support intensive farming practices. The primary crops cultivated in the area include cereals and industrial crops such as wheat, barley, maize, sugar beet, sunflower, and alfalfa. Although dryland farming continues in certain parts of the region, the expansion of irrigation infrastructure has contributed to a notable increase in irrigated agricultural practices. With these characteristics, Hotamış represents a significant example of the coexistence of both traditional and modern agricultural systems in Turkey. Land consolidation and in-field development projects have also been implemented across Hotamış and its surrounding areas. As part of these initiatives, agricultural parcels have been reorganized into geometrically larger and more uniform fields with well-defined and regular boundaries. This restructuring provides a valuable advantage for remote sensing applications, particularly in segmentation and classification tasks based on satellite imagery. The homogeneity and structural clarity of the parcels enhance the potential for accurate spatial analysis, making the region a suitable case study for agricultural boundary delineation research.

2.2. Sentinel-2 imagery

Sentinel-2 is an optical satellite mission developed by the European Space Agency (ESA) under the Copernicus Programme of the European Commission, designed to support sustainable Earth observation and environmental monitoring efforts. The constellation consists of three satellites—Sentinel-2A, Sentinel-2B, and Sentinel-2C—launched in

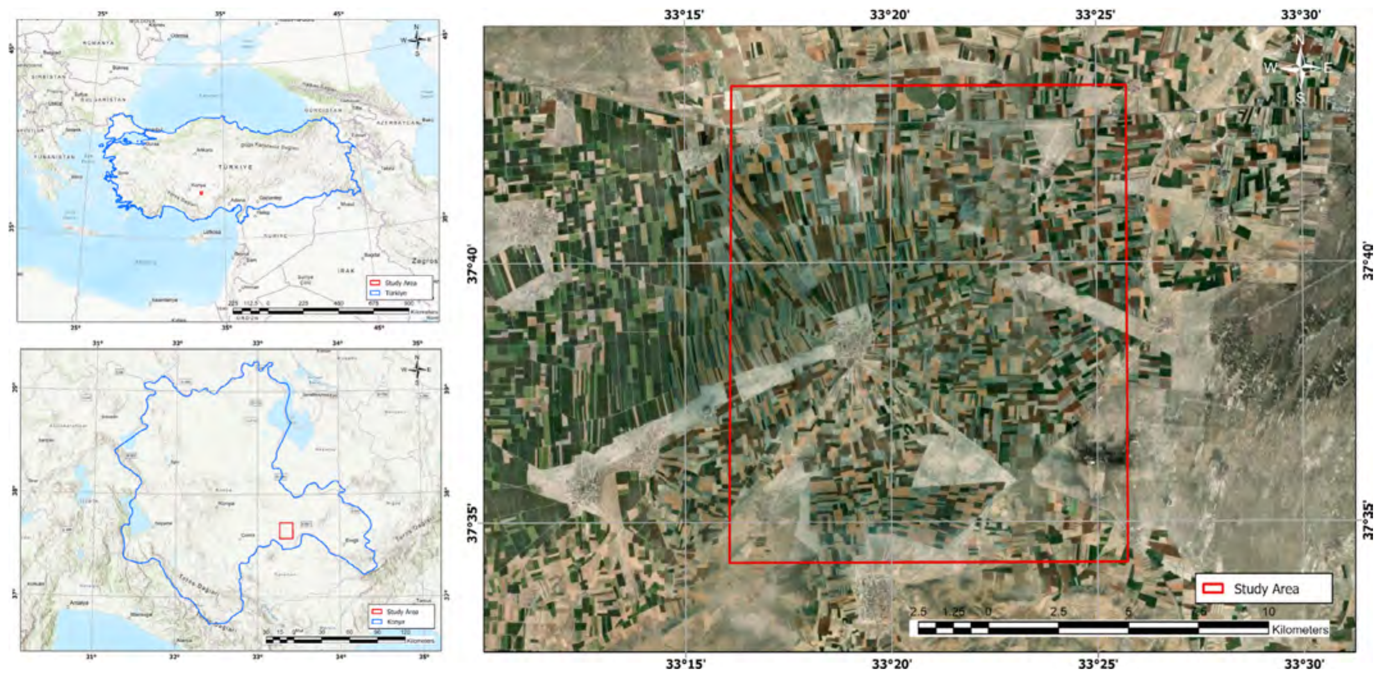

Fig. 1. Study area.

June 2015, March 2017, and September 2024, respectively (Table 1). Each satellite is equipped with a 13-band multispectral instrument capable of capturing data at spatial resolutions of 10, 20, and 60 m, with a radiometric resolution of 12 bits (Li and Roy, 2017). The Sentinel-2 constellation plays a pivotal role in a wide range of environmental and agricultural applications, including land cover change detection, vegetation health assessment, crop monitoring, and natural disaster surveillance. Its high revisit frequency, broad swath width (290 km), and rich spectral configuration make it particularly valuable for continuous, large-scale, and temporally dense observations. In agricultural research, Sentinel-2 data are widely utilized for tracking crop development and assessing phenological stages at the parcel level, particularly through the calculation of vegetation indices such as the Normalized Difference Vegetation Index (NDVI) and similar spectral metrics. These capabilities have made Sentinel-2 a cornerstone data source for both operational and research-oriented applications in precision agriculture, land use planning, and environmental monitoring.

2.3. PlanetScope imagery

PlanetScope is one of the world's largest commercial Earth observation systems, consisting of a constellation of small satellites commonly referred to as "Doves," operated by Planet Labs. Characterized by its high temporal resolution, PlanetScope is capable of imaging the entire Earth's land surface on a near-daily basis (Mansaray et al., 2021). The system has evolved through three main satellite generations. The first generation, Dove (PS2), launched in 2013, provided tri-band imagery composed of red, green, and blue spectral bands. The second generation, Dove-R (PS2.SD), launched in 2018, introduced a four-band configuration with the addition of the near-infrared (NIR) band. Since 2020, the third and current generation, SuperDove (PSB.SD), has been operational. SuperDove satellites offer enhanced capabilities through an expanded spectral range of eight bands, including blue, green, red, NIR, red-edge, and additional vegetation-sensitive wavelengths (Planet, 2022).

This broader spectral coverage, which is partially aligned with Sentinel-2′s band structure, enhances their applicability for agricultural monitoring and time-series analyses. PlanetScope imagery has a spatial resolution ranging between approximately 3.0 and 3.7 m, with a revisit frequency that enables near-daily temporal resolution (Table 2). These characteristics make it particularly well-suited for high-precision monitoring in smallholder agricultural landscapes. In comparison, Sentinel-2 satellites—while widely used for agricultural monitoring—provide imagery at a

Table 1
Sentinel-2 specifications.

Mission characteristic	Sentinel-2	Mission CHARACTERISTIC	Sentinel-2
Orbit altitude	786 km	Pixel size (orthorectified)	10/20/60 m
Sensor type	Coastal Aerosol, Blue, Green, Red, Red Edge (x3), NIR, SWIR (x2), Water Vapor, Cirrus	Bit depth	12 bit
Frame size	100 km × 100 km	Revisit time	3–5 days

Table 2
PlanetScope PSB. SD specifications.

Mission characteristic	PSB. SD	Mission characteristic	PSB. SD
Orbit altitude	475–525 km	Pixel size (orthorectified)	3 m
Sensor type	Coastal Blue, Blue, Green I, Green II, Yellow, Red, Red-edge, and NIR	Bit depth	Visual: 8-bitAnalytic Rad.:16 bitAnalytic SR 16-bit
Frame size	32.5 km × 19.6 km	Revisit time	Daily at nadir

spatial resolution of 10 m. This level of resolution may be insufficient for reliably analyzing small parcels.

According to Vajsová et al. (2020), Sentinel-2 data yield accurate results only when an agricultural parcel contains at least eight complete pixels, and when pixel loss remains below 60 %. This translates to a minimum parcel size of approximately 0.8 ha (8,000 m^2). Below this threshold, the reliability of analysis decreases markedly. Given that nearly 35 % of the parcels in the study area are classified as small (<2.56 ha) or very small (<0.64 ha), and that approximately 900 parcels fall below 1 ha, PlanetScope (PSB. SD) imagery was preferred in this study. The use of higher-resolution data facilitated more precise and detailed delineation of agricultural parcel boundaries following segmentation. A comparison of the spectral band configurations across the three generations of PlanetScope satellites and Sentinel-2 is illustrated in Fig. 2.

2.4. Reference parcels (agriculture parcel) data

To evaluate both the quantitative and qualitative accuracy of the agricultural parcel boundaries generated using the SAM model, official reference parcels were employed as ground truth data (Fig. 3). An agricultural parcel is defined as a land unit owned or managed by the same individual(s), with clearly defined boundaries, and utilized for agricultural production (Ministry of Agriculture and Forestry, 2018). As part of the Agricultural Production Registration System (TÜKAS) project, implemented by the Republic of Turkey Ministry of Agriculture and Forestry, reference agricultural parcels were delineated by digitizing areas under cultivation or with agricultural potential within cadastral parcel boundaries. This process utilized orthophoto imagery with 50 cm spatial resolution and high-resolution satellite images (TÜKAS, 2020). During

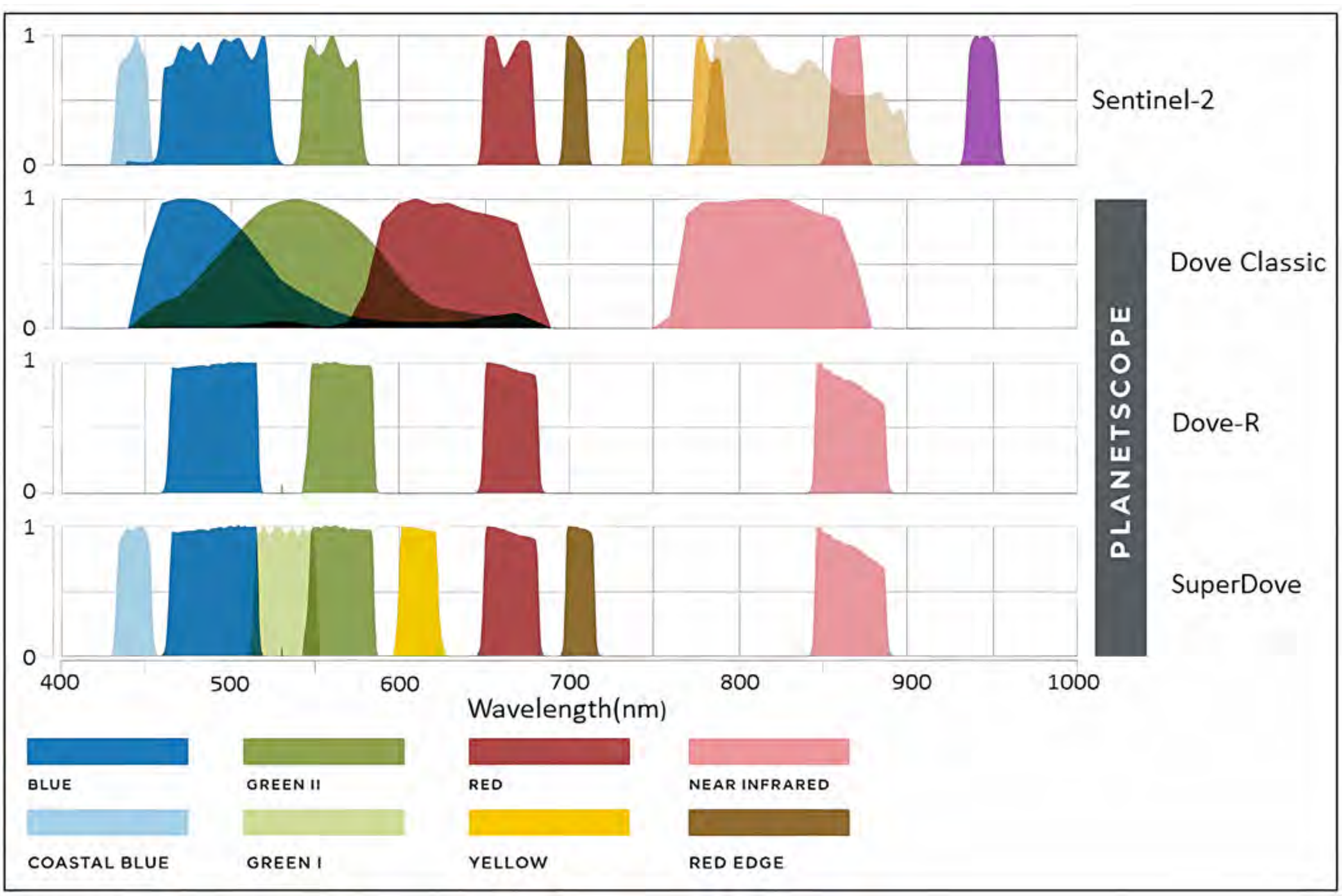

Fig. 2. Spectral bands of Planet's Dove Classic, Dove-R, and SuperDove satellites compared to Sentinel-2 (Planet 2024).

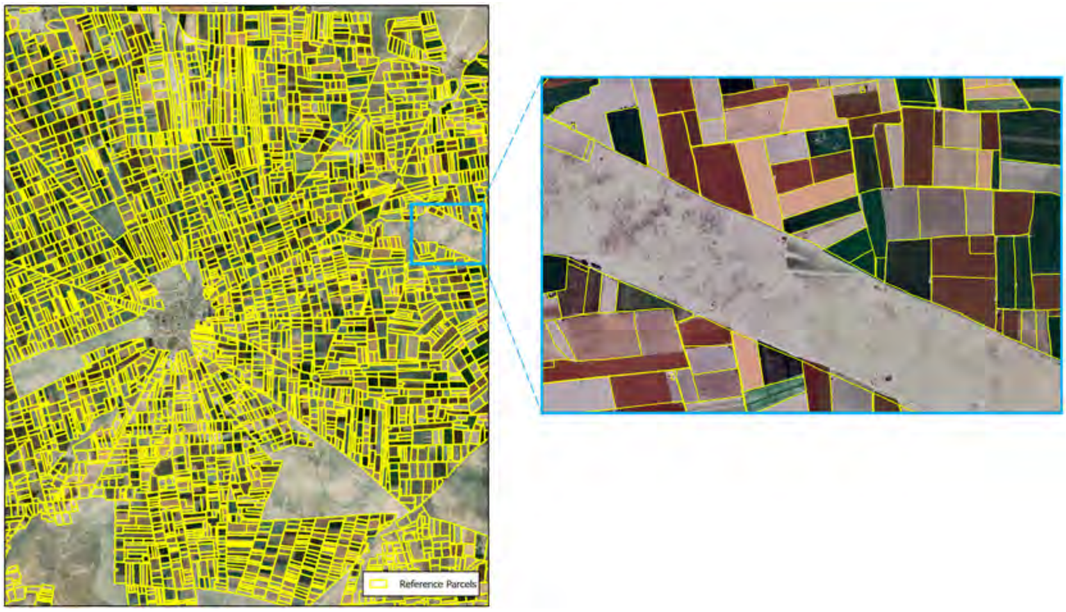

Fig. 3. Reference parcels within the study area.

digitization, non-agricultural features such as residences, barns, abandoned buildings, water bodies, and rocky patches were excluded, ensuring that only active agricultural land was represented. Since 2013, these digitized parcels have served as the foundational dataset for the Farmer Registration System (FLSS), agricultural subsidies, and various administrative applications (FDP Regulation, 2018). However, the actual usage boundaries of agricultural parcels may differ from their digitized legal boundaries due to the dynamic and seasonal nature of agricultural practices. For instance, the cultivation of multiple crops at different phenological stages within the same parcel, partial planting of a field, or double cropping within a single season may not alter the legal extent of a parcel but can lead to substantial visual variations in texture, color, and pattern within the satellite imagery (Demir and Yomralıoğlu, 2020). Although the reference parcels represent areas dedicated exclusively to agricultural use, these temporal and structural variations introduce discrepancies between legal boundaries and actual crop extent. Therefore, such variability must be considered when assessing the accuracy of segmentation-based parcel delineation models. In this study, reference parcel boundaries were overlaid with PlanetScope (PSB.SD) imagery acquired on three phenologically representative dates, and their temporal validity was examined through visual interpretation. According to Lesiv et al. (2019), agricultural parcels are classified by size as follows: extra-large (>100 ha), large (16–100 ha), medium (2.56–16 ha), small (0.64–2.56 ha), and very small (<0.64 ha). In total, 4212 reference parcels were used in the study area. Among these, 58 were categorized as large, 2672 as medium, 1374 as small, and 108 as very small parcels. The average parcel size across the study area was calculated to be approximately 4 ha.

2.5. Farmer declaration parcels data

The Farmer Registration System (FRS) is a national database managed by the Ministry of Agriculture and Forestry of the Republic of Turkey, which records production data of individuals and legal entities engaged in agricultural activities throughout the country. The primary objective of the system is to ensure that agricultural subsidies are distributed effectively, transparently, and in an auditable manner by collecting parcel-based production declarations from farmers (Şimşek and Durduran, 2023). In each production season, farmers submit applications to the system by providing documentation of their ownership or usage rights for the cultivated parcels, alongside declarations of their production practices. The FRS database contains not only geometric information on agricultural parcels but also extensive qualitative attributes, including province, district, neighborhood, island-parcel number, agricultural parcel number, crop type, cultivated area, surface area, cadastral area, as well as planting and harvesting dates (Şimşek, 2024). However, since the data is largely based on farmer self-declarations and requires seasonal updates, its use as reference data in remote sensing and digital mapping studies may be limited. Therefore, the seasonal variability and the dynamic nature of the system must be carefully considered during accuracy assessments (Şimşek, 2025). Within the study area, the most frequently declared crops in the FRS include wheat, maize, sunflower, alfalfa, sugar beet, and lentil (Fig. 4). For each crop, approximately 100 parcels were selected, ensuring homogeneous spatial distribution throughout the region. These parcels were overlaid with NDVI time series data derived from Sentinel-2 satellite

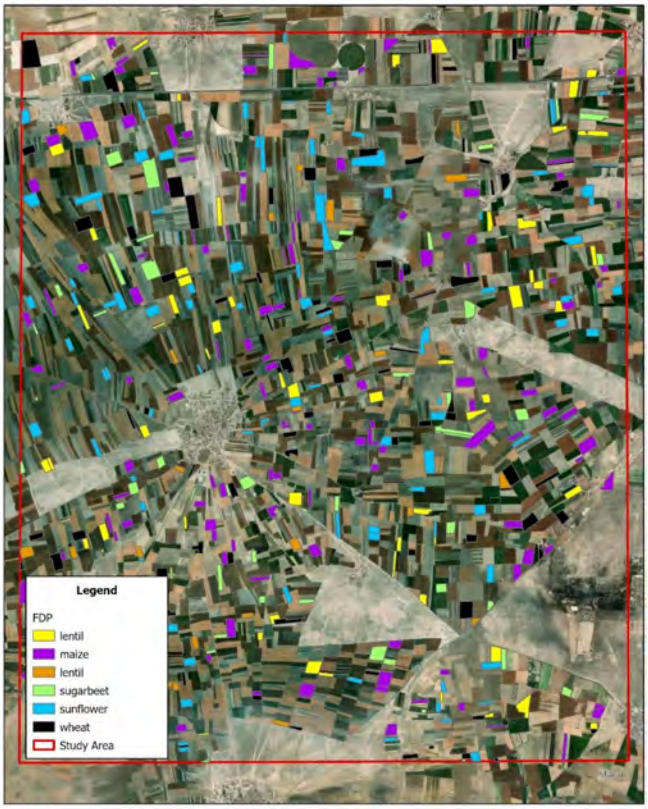

Fig. 4. Farmer declaration parcels within the study area.

imagery to generate spectral decomposition curves for each individual plot. Subsequently, by averaging these curves across all plots within each crop class, characteristic spectral separation profiles were established to distinguish the crops based on their phenological and spectral signatures.

3. Methods

3.1. General workflow overview

This study proposes a comprehensive workflow for agricultural parcel segmentation leveraging satellite imagery and the Segment Anything Model (SAM)**.** The methodology is structured into four primary stages (Fig. 5). In the first stage, raster data (PlanetScope and Sentinel-2) and vector data (Farmer Declaration Parcels and Reference Parcels) are obtained for the analysis. In the second stage, the obtained raster and vector data w e r e subjected to various preprocessing steps before SAM segmentation. In this context, phenological stages of the crops in the study area were determined by analyzing Sentinel-2 NDVI images, and PlanetScope images to be used for segmentation were determined with reference to these stages. In the third stage, the hyper-parameters of the SAM model were optimized and the segmentation process was applied for each data set. After segmentation, various post processing operations were performed to ensure that only the areas corresponding to agricultural parcels remained. In the fourth and final stage, the segmentation results were compared with the reference parcels to analyze the accuracy and the findings were discussed.

3.2. Determination of phenological stages and selection of time series PlanetScope images

The spectral reflectance characteristics of agricultural lands exhibit significant temporal variation corresponding to the phenological stages of crops. These stages typically encompass pre-sowing, early growth, mid-growth, flowering, full canopy development, pre-harvest, and harvest periods. Such spectral variability between phenological phases suggests that satellite imagery acquired at different times can more effectively and accurately delineate field boundaries according to crop types (Tripathy et al., 2024). This is especially advantageous in regions where crops have diverse planting dates and developmental cycles, as phenology-driven spectral changes facilitate enhanced boundary detection. In this study, to improve the accuracy and reliability of parcel boundary delineation, the phenological calendar of the main crop species commonly cultivated in the study area was obtained from the relevant District Directorate of Agriculture (Fig. 6). Additionally, 15-day interval NDVI time series were derived from Sentinel-2 Level 2A satellite imagery using the Google Earth Engine (GEE) platform. NDVI values for each crop type were calculated by overlaying these images with crop-specific farmer declaration parcels. Spectral separation profiles for each crop were generated using the NDVI time series, with anomalous values excluded to ensure representative averages. These mean spectral curves were then compared against the phenological calendar data to assess compatibility. This comparative analysis, together with the phenology-sensitive temporal resolution of the NDVI data, informed the selection of optimal imagery dates for more spatially precise and detailed parcel boundary delineation.

Examination of the NDVI time series curves presented in Fig. 7 reveals distinct spectral separations for each crop type throughout the growing season. Winter crops such as wheat and lentils exhibit vegetative growth approximately between the 110th and 215th day of the year, whereas summer crops like maize and sunflower show active growth phases roughly from the 155th to the 290th day. In contrast, sugar beet demonstrates an extended developmental period lasting up to around the 320th day. The perennial forage crop alfalfa displays multiple sowing and harvesting cycles annually, reflected by pronounced fluctuations in its NDVI curve.

These spectral separation profiles identify temporal windows during which crops exhibit maximum spectral divergence due to phenological differences. Consequently, it is predicted that agricultural parcels with differing crop types can be more clearly and distinctly delineated during the periods of highest spectral separation, specifically on May 25, July 19, and September 22, 2022 (Fig. 7). Cloud-free

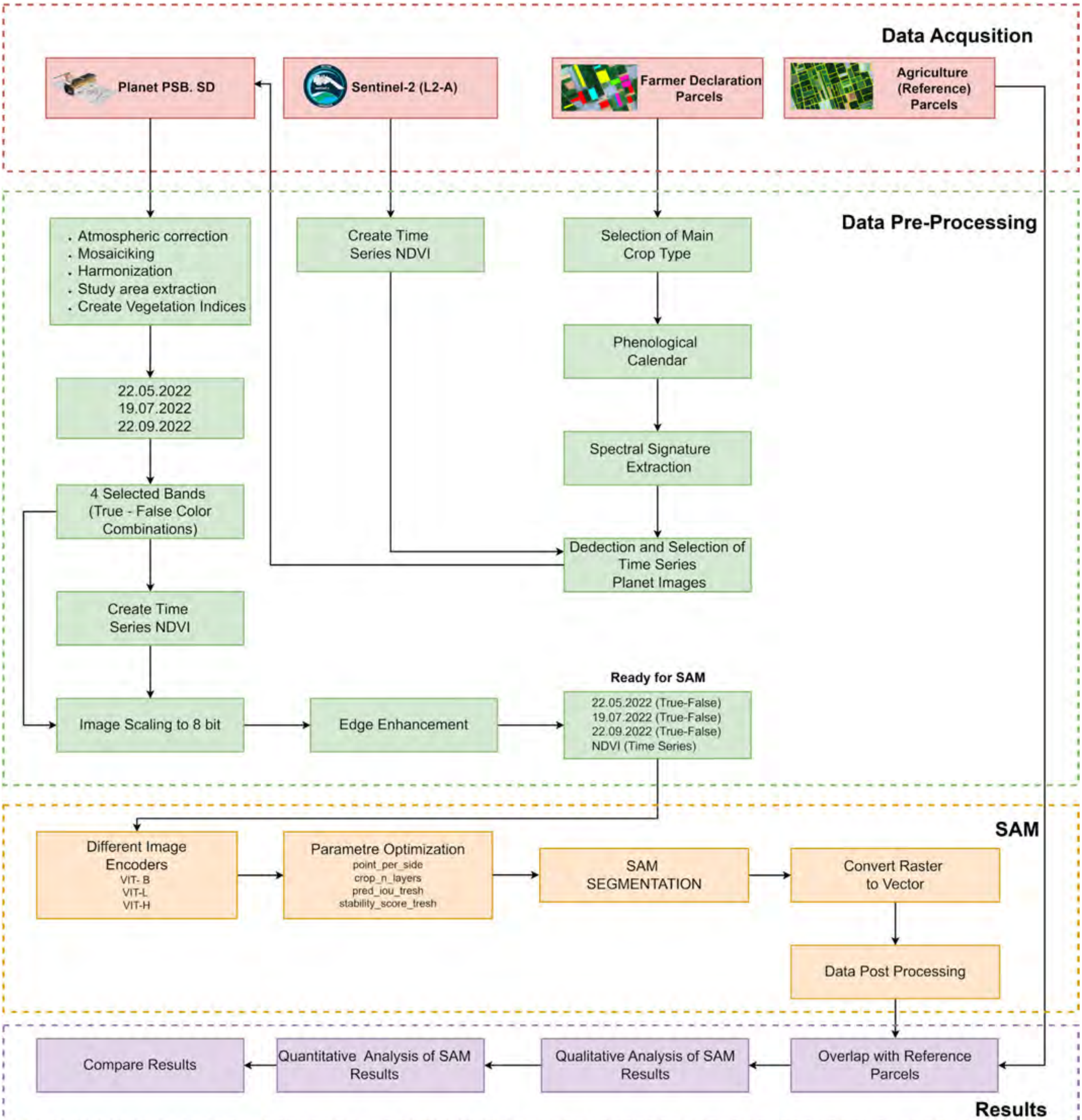

Fig. 5. General workflow of this study.

PlanetScope (PSB.SD) images corresponding to the relevant dates were acquired. Since PlanetScope imagery provides a daily acquisition frequency, scenes affected by cloud coverage did not pose any issue during the data selection process. When higher spatial detail is desired for precise parcel boundary delineation, super-resolved images generated from Sentinel-2 data (with a 5-day revisit cycle) can be used as an alternative data source. This phenology-driven approach leverages the unique spectral responses of crops at different developmental stages to guide the selection of satellite imagery dates most suitable for segmentation. Moreover, it mitigates the substantial limitations associated with processing large volumes of high-resolution images by restricting the analysis to only the most informative dates. This targeted selection reduces data redundancy and significantly enhances processing efficiency, enabling precise agricultural boundary delineation with optimized resource use.

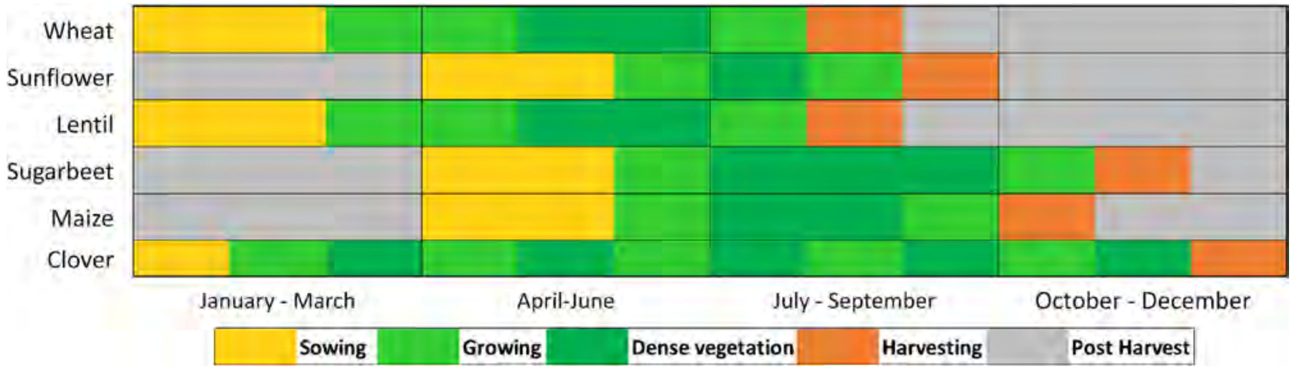

Fig. 6. Crop calendar in the study area.

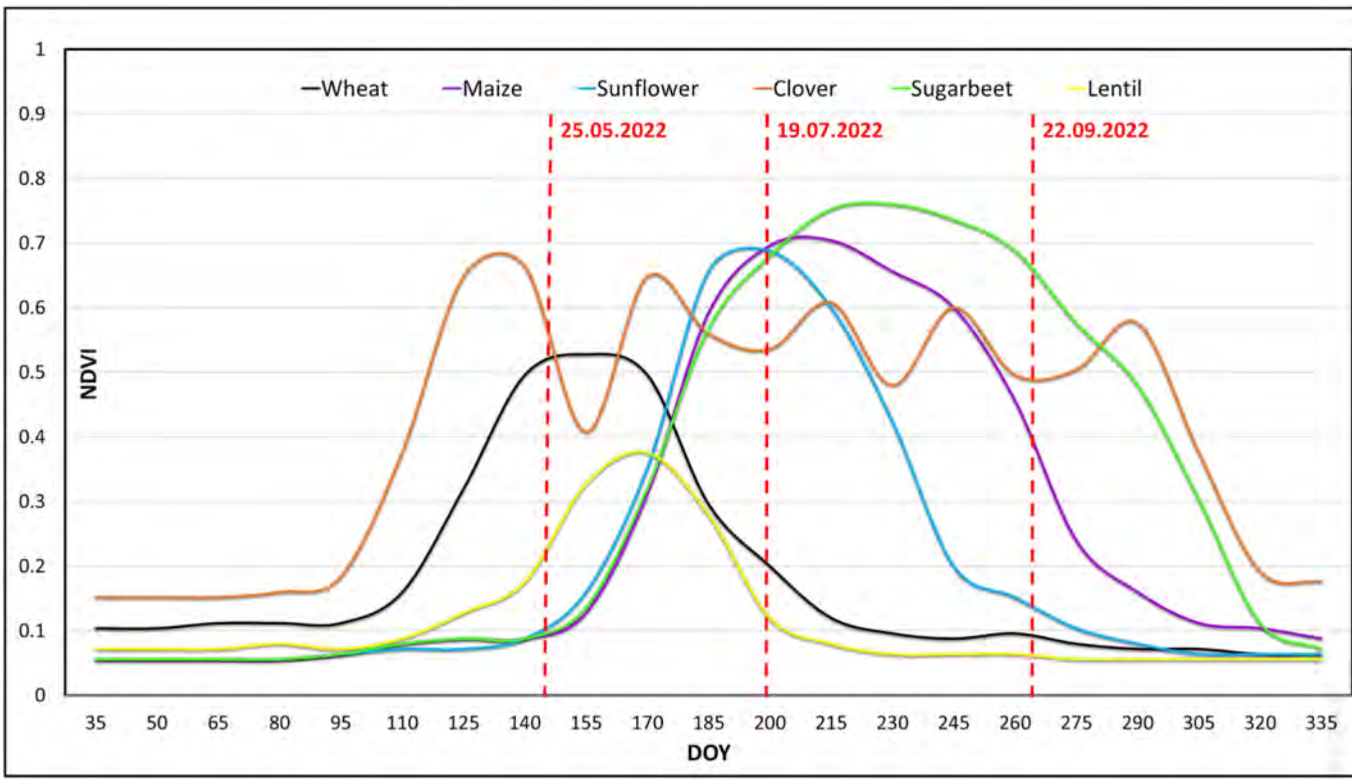

Fig. 7. NDVI spectral separation curves for plants.

3.3. Pre-processing of PlanetScope

In this study, agricultural parcels were automatically detected using the SAM with 8-band data obtained from the PSB.SD sensor of the PlanetScope (PS) satellite constellation. PlanetScope imagery is provided with two distinct geometric correction levels: basic and ortho. While Basic scenes are intended for users requiring advanced pre-processing and are not orthorectified nor terrain-corrected, Ortho scenes are georeferenced and orthorectified products suitable for direct analysis (Şahin and Kaplan, 2023). The data employed in this research corresponds to the PS Ortho Analytic 8B SR (Level 3B) product, which includes scenes orthorectified with a positional accuracy of less than 10 m root mean square error (RMSE), projected using the UTM/WGS84 coordinate reference system (Mudereri et al., 2019). These scenes were atmospherically corrected to surface reflectance (SR) level using the 6SV radiative-transfer model, which renders them appropriate for quantitative remote-sensing applications. To ensure spectral consistency across multiple generations of PlanetScope sensors (PS2, PS2.SD and PSB.SD), Planet provides a normalization and harmonization tool (Vermote et al., 1997; Wilson, 2013). This tool transforms older-sensor surface-reflectance values to align with those of the PSB.SD platform and with the spectral response standard of Sentinel–2 (Planet, 2023a). The harmonization workflow comprises several key steps: (i) utilization of 6SV-corrected SR imagery as the starting point; (ii) per-band linear-regression modeling between cloud-free PlanetScope pixels and time-matched Sentinel-2 reference data, thereby deriving conversion coefficients for each band; (iii) prioritisation of pseudo-invariant features (PIFs) stable, low-reflectance surfaces that minimize BRDF and atmospheric variation as core training samples; (iv) comprehensive cloud/shadow masking to exclude unreliable pixels from the regression; and (v) inter-sensor spectral-response system (SRS) mismatch correction, ensuring images from older sensors (PS2, PS2.SD) are brought into compatibility with PSB.SD/Sentinel-2 baselines. Together, these stages form an operational yet approximate alignment procedure that significantly reduces inter-scene radiometric inconsistencies while preserving within-scene spatial and temporal variability. As a result, multi-generation PlanetScope scenes become sufficiently consistent for comparative analyses, multi-temporal change detection, and seamless integration with Sentinel-2 time series (Planet, 2023b).After harmonization, all individual PSB.SD e scenes were mosaicked to ensure seamless spatial coverage and then clipped to match the boundaries of the designated study area.

The SAM-based zero-shot segmentation process was realized through the open-source Python package SAM-Geo, which enables the segmentation of geospatial data using SAM. The SAMGeo library allows the direct application of the SAM algorithm on raster images; however, this process is limited to 3-band and 8-bit resolution remote sensing images (Huang et al., 2024). For this reason, the PlanetScope (PSB.SD) images of three different dates (25.05.2022–19.07.2022–22.09.2022) used in the study were converted into 3-band format with true color (B6-B4-B2) and false color (B8-B6-B4) band combinations and then converted to 8-bit format to be suitable for SAMGeo segmentation.

3.4. Multi temporal data

Although true color and false color composites provide valuable information for interpreting parcel boundaries in automatic delineation studies, they may be limited in capturing the phenological dynamics of vegetation. In this context, the use of vegetation indices such as the Normalized Difference Vegetation Index (NDVI), which effectively reflect the temporal and spatial variability of vegetation development, offers considerable advantages for improving segmentation accuracy. NDVI quantitatively assesses plant health and canopy density by exploiting the difference in reflectance between the red and near-infrared spectral bands. This enhances spectral contrasts between different crop types and phenological stages, making the boundaries between parcels more pronounced. Agricultural landscapes exhibit substantial spectral variation throughout the crop growth cycle. Phenological phases such as pre-planting, early growth, and maturity correspond to distinct NDVI values that directly influence the clarity of field boundaries. For example, during early development, parcel edges are more discernible as crops have yet to fully cover the field, whereas in the maturity phase, dense vegetation may obscure boundaries, especially when adjacent plots contain the same crop species. Conversely, where neighboring parcels host different crops, phenological progression often increases boundary visibility due to spectral dissimilarity.

Satellite-based time series analyses provide rich temporal and spectral information, facilitating the discrimination of in-field and out-field areas as well as fixed plot edges (Graesser and Ramankutty, 2017; Tetteh et al., 2021; Cai et al., 2023; Song, 2023). Consequently, combining NDVI images acquired at dates representing different phenological stages enables the analysis of parcel boundaries at their most distinct temporal signatures, thereby significantly enhancing segmentation performance. This approach also improves boundary delineation in plots with multiple crop phenologies or partial crop cover within the same growing season.

Utilizing multi-temporal NDVI time series data is thus an effective strategy to increase the precision of agricultural parcel boundary detection. Literature evidence that integrating segmentation results from multiple dates substantially improves overall detection accuracy. For instance, Tripathy et al. (2024) employed 2-meter resolution SkySat imagery to detect agricultural parcels using the Segment Anything Model (SAM), comparing segmentation accuracies obtained by combining images from four distinct dates. Their findings indicate a roughly 25 % increase in detection accuracy when multi-date segmentations were fused. This improvement stems from the fact that some parcel boundaries are more distinguishable during early growth stages, while others become clearer at later developmental phases. Therefore, multi-temporal analysis provides more robust and adaptable results compared to single-date segmentation.

NDVI time series-based approaches facilitate more detailed and accurate segmentation of agricultural landscapes by capturing periods characterized by high spectral homogeneity within parcels and pronounced heterogeneity between parcels (Zhang et al., 2020). In the present study, NDVI images were generated for the PlanetScope (PSB.SD) scenes acquired on May 25, July 19, and September 22, 2022 (Fig. 8). Monthly NDVI images corresponding to the selected dates were combined using the band stacking method, thereby generating a multi-temporal NDVI Time Series (NDVI-TS) composite. This integrated dataset consolidates spectral information from different phenological stages within a common spatial framework, enabling a more consistent representation of surface characteristics. The resulting NDVI-TS composite was formatted as a 3-band, 8-bit image in accordance with the input requirements of the SAM and subsequently employed as a primary input in the segmentation analyses.

3.5. Histogram equalization and stretching for edge enhancement

Feature engineering is a crucial step in machine learning, transforming raw data into more meaningful, discriminative, and informative inputs that enhance model

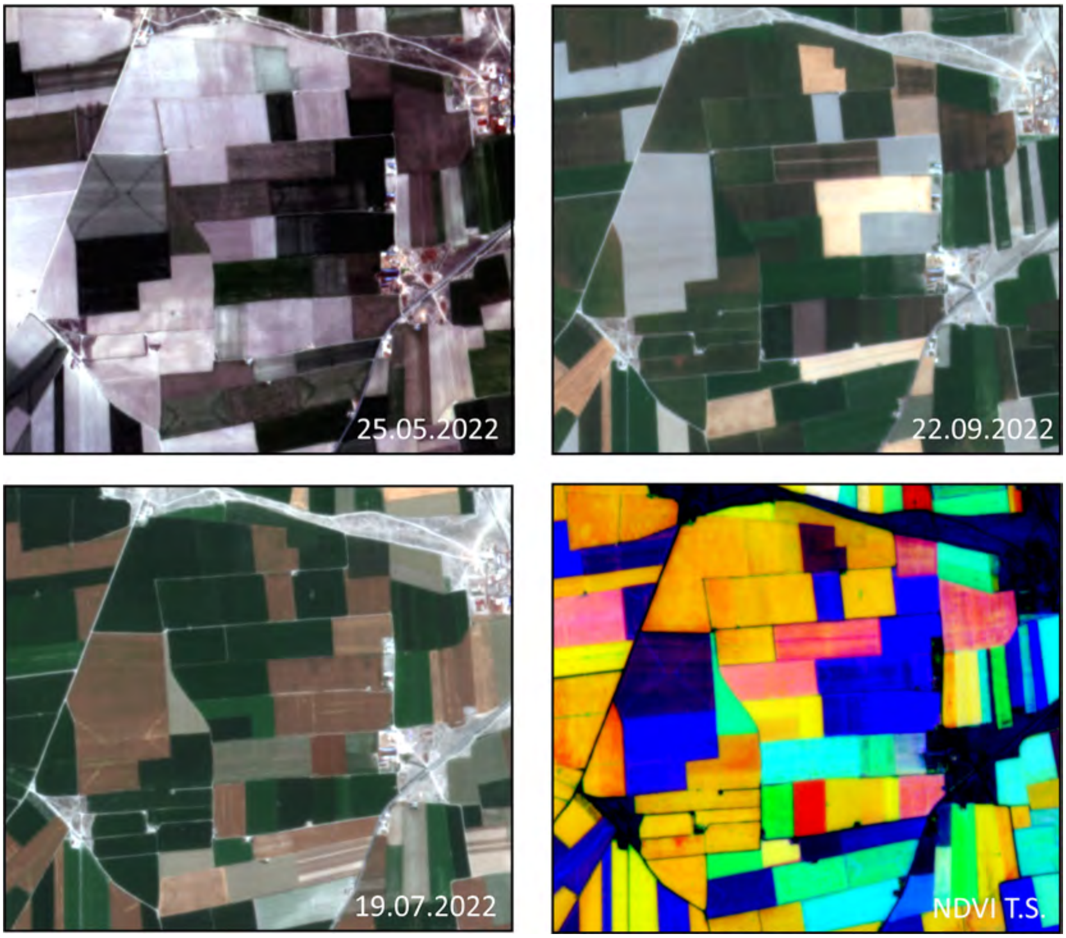

Fig. 8. PlanetScope (PSB. SD) and NDVI TS imagery.

performance (Qiu et al., 2016). This process is commonly employed to improve both the accuracy and generalization capabilities of models. However, there remains no clear consensus in the literature regarding the operational impact of feature engineering techniques specifically within remote sensing workflows (Li and Gong, 2020). Nevertheless, applying edge enhancement prior to segmentation—particularly in satellite imagery—has been widely recommended to better delineate linear features and object boundaries (Chen et al., 2024).

Contrast enhancement methods such as histogram equalization and histogram stretching are frequently used for this purpose, expanding the dynamic range of images and making structural details in low-contrast regions more prominent (Gonzalez and Woods, 2018). The positive influence of these preprocessing techniques on segmentation accuracy has been demonstrated in multiple studies. For example, Tripathy et al. (2024) reported a 3 % increase in segmentation accuracy when applying edge enhancement via histogram equalization during agricultural area detection using the Segment Anything Model (SAM). This underscores the value of contrast enhancement in improving class separability, particularly in agricultural contexts where spectrally similar classes with ambiguous boundaries are common.

Similarly, Graesser and Ramankutty (2017) employed multiscale contrast-limited adaptive histogram equalization combined with adaptive thresholding on Landsat Surface Reflectance Climate Data Record (CDR) datasets to identify agricultural regions in South America, achieving notable improvements in segmentation quality. Zhang et al. (2023) further developed a deep learning-based method emphasizing edge enhancement and demonstrated substantial segmentation accuracy gains, especially when data availability is limited. Collectively, these findings suggest that image contrast optimization during preprocessing significantly enhances the performance of machine learning-based segmentation models by sharpening object boundaries, particularly in high-resolution remote sensing data (Gupta et al., 2023).

In this study, to enhance the segmentation performance of the SAM algorithm, contrast enhancement procedures were applied to true color, false color, and NDVI time series images from three different dates prior to segmentation. Specifically, histogram equalization and histogram stretching techniques were utilized to broaden the images' dynamic ranges, while edge sharpening methods accentuated object boundaries (Fig. 9). These preprocessing steps contributed to clearer delineation of agricultural patterns and parcel boundaries, particularly in low-contrast areas,

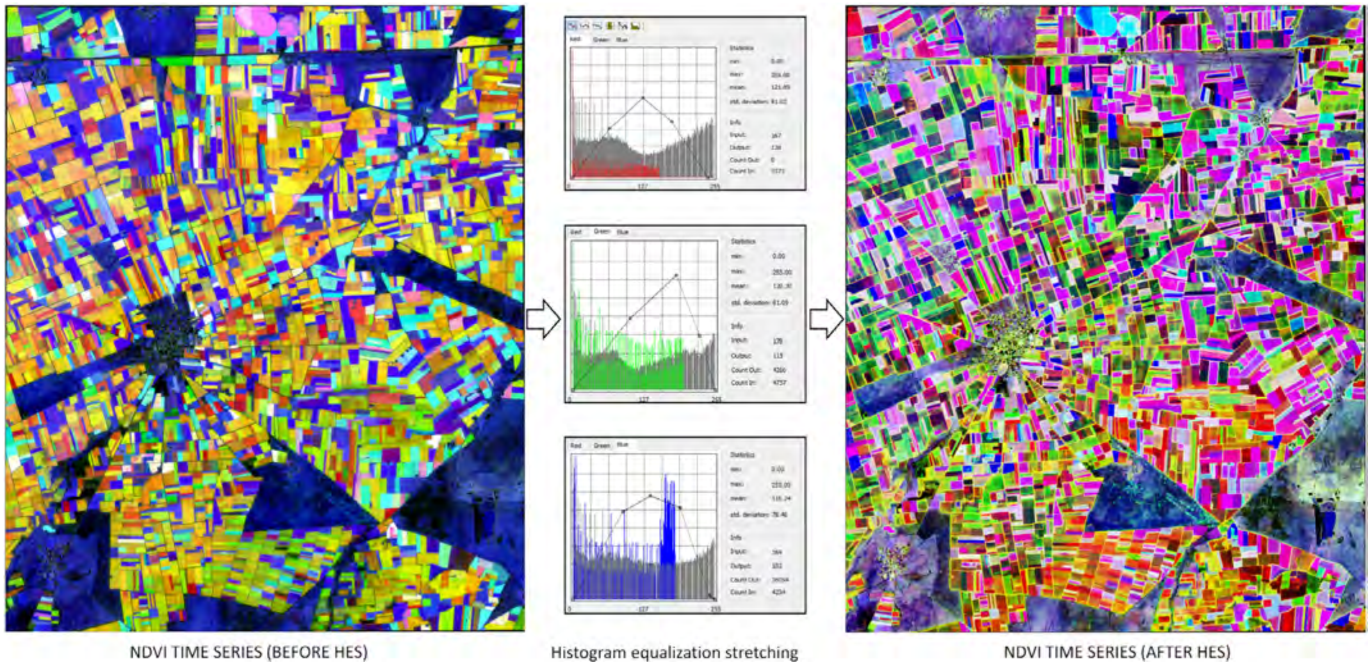

Fig. 9. PlanetScope (PSB. SD) and NDVI TS.

enabling the SAM model to produce more accurate and detailed segmentation results.

3.6. SAM (Segment Anything Model)

Recent developments in the field of computer vision have led to the emergence of groundbreaking models, especially in visual segmentation tasks. One of these models, SAM, is a zero-shot artificial intelligence model developed by Meta AI and trained for general-purpose visual segmentation tasks (Kirillov et al., 2023). SAM is capable of determining object boundaries on an image with high accuracy based on a prompt provided by the user (e.g. a point, rectangle or freehand drawing). This distinguishes it from traditional deep learning-based segmentation models that are trained only for a specific class or application (Osco et al., 2023). SAM provides a pre-trained, general-purpose segmentation framework for a wide range of applications. The architecture of the model consists of three main components: image encoder, prompt encoder and mask generator. The image encoder transforms high-resolution images into vector space with a Vision Transformer (ViT) based structure, while the prompt encoder encodes the input from the user (e.g. a dot or a box). The mask generator combines these two components to produce the segmentation mask (Kirillov et al., 2023; Li et al., 2024). In the training phase, SAM is trained with a large dataset called "SA-1B" consisting of 11 million segmentation masks, making it highly robust to object diversity. Thanks to this powerful infrastructure, SAM is able to perform highly accurate segmentation on different data types without any retraining, even in zero-shot or few-shot scenarios (Zhou et al., 2023). In this study, high spatial resolution (3 m) PlanetScope SuperDove satellite imagery was employed to analyze the segmentation performance of agricultural fields in detail. As SAM is highly sensitive to spatial resolution for accurately delineating object boundaries, PlanetScope SuperDove data featuring advanced radiometric calibration and geometric corrections was preferred to achieve more precise and reliable delineation of agricultural parcel boundaries.

3.7. SAM for agricultural field boundary delineation

The delineation of agricultural parcel boundaries has long been a key research focus in remote sensing and geographic information systems (GIS) applications. Over time, the methodologies developed for this purpose have evolved significantly—from early approaches relying on manual digitization to advanced deep learning-based methods, and more recently to the use of large-scale visual models known as foundation models. In the early 2010s, agricultural parcel boundaries were primarily extracted through manual interpretation or classical image processing techniques such as edge detection and region growing. However, these conventional methods often suffered from limited accuracy and poor generalizability, particularly in landscapes characterized by heterogeneous and small-scale parcel structures. To address these challenges, deep learning models began to be widely adopted after 2016. For instance, Waldner and Diakogiannis (2019) demonstrated that agricultural field boundaries could be accurately delineated using the ResUNet-a architecture with Sentinel-2 satellite imagery. Building on this, Hossen et al. (2025) proposed the use of transfer learning and weakly supervised learning approaches to overcome data scarcity problems specific to smallholder agricultural systems, enabling more scalable and adaptable solutions. More recently, the introduction of the Segment Anything Model (SAM) by Meta AI has brought about a paradigm shift in image segmentation. SAM is capable of performing segmentation tasks without requiring class-specific training or dense annotations, making it especially suitable for use cases with limited or no labeled data. Tripathy et al. (2024) applied SAM on 2-meter resolution SkySat imagery and reported an accuracy of approximately 58 % in delineating agricultural field boundaries in smallholder farming areas. In a separate study, Xie et al. (2025) proposed a hybrid model called fab-SAM by integrating SAM with DeepLabV3+, which yielded a 23.5 % and 15.1 % improvement in Intersection over Union (IoU) scores across two benchmark datasets, respectively. Building on these advances, the present study aims to evaluate the potential of the SAM model for automatic segmentation of agricultural parcel boundaries using high-resolution single-date and multi-temporal PlanetScope (PSB.SD) satellite imagery. Three distinct acquisition dates were selected to represent different phenological stages of crop development. For each date, natural color composites, false color composites, and NDVI time series (NDVI TS) images were generated to provide spectrally enriched inputs to the SAM model. The segmentation results obtained from these different inputs were then systematically compared to assess the model's performance under varying temporal and spectral conditions.

3.8. SAM tunable parameters

SAM, is a general-purpose deep learning framework for image segmentation that incorporates three different Vision Transformer (ViT)-based encoder variants and a range of adjustable hyperparameters. These encoders include ViT-B (91 million parameters), ViT-L (308 million parameters), and ViT-H (636 million parameters). The ViT-B variant, representing the smallest and most computationally efficient version of the model family, operates with lower latency and reduced resource demand. However, this computational advantage comes at the cost of reduced segmentation precision in complex scenarios. On the other hand, the ViT-H model, with the highest number of parameters, offers superior segmentation performance, particularly in detailed and heterogeneous landscapes. Yet, this accuracy gain requires significantly more GPU memory and processing time. As a result, while ViT-B may be suitable for systems without GPU acceleration or in lightweight applications, ViT-L and ViT-H are more appropriate for high-resolution agricultural mapping tasks, especially those involving fine-scale parcel delineation. To effectively leverage SAM in agricultural segmentation applications, careful optimization of its hyperparameters is essential. In addition to segmentation accuracy, processing efficiency is a critical factor—particularly for large-scale or operational implementations requiring the automated extraction of parcel boundaries across extensive agricultural landscapes. Therefore, balancing model complexity with computational constraints is vital for ensuring both the feasibility and accuracy of the segmentation process. In this context, the effects of key SAM hyperparameters on segmentation performance are summarized in Table 3. This comparative overview provides a reference for researchers to guide the selection of model configurations suitable for their specific remote sensing applications, considering both technical capacities and expected output quality.

The accuracy and spatial precision of the segmentation outputs produced by the SAM model are highly dependent on the optimization of certain key hyperparameters. Among these, points_per_side and crop_n_layers stand out as particularly influential in determining segmentation quality. The points_per_side parameter defines the sampling density of input prompts across the image, from which the segmentation masks are derived. Increasing this value enhances spatial granularity and object delineation by enabling the model to sample a larger number of regions. However, this improvement comes at the cost of increased computational load and processing time. Similarly, the crop_n_layers parameter enables a multi-scale segmentation strategy by dividing the input image into different hierarchical layers. Higher values of this parameter facilitate the accurate detection of small or geometrically complex objects through refined spatial analysis. Nonetheless, each additional cropping layer intensifies

Table 3
SAM parameters.

Parameter	Value range	Description	Default value
Points_per_side	16–512	Specifies the number of sampling points placed along each side of the input image. Higher values increase segmentation detail but also processing time	32
Crop_n_layers	1–10	Determines how many times the segmentation is repeated at different resolution levels. More layers yield finer results at the cost of longer computation	0
Pred_iou_thresh	0.5–1	IoU (Intersection over Union) threshold for accepting predicted masks. Lower thresholds increase speed but may lead to more false positives	0.88
Stability_score_thresh	0.5–1	A measure of the stability of segmentation masks. Lower values generate more segments, but may reduce consistency.	0.95
Crop_n_points downscale factor	1–5	A scaling factor used to reduce sampling density in deeper crop layers. Higher values improve speed but may cause loss of detail.	1
Min_mask Region area	10–1000	Specifies the minimum area (in pixels) for an accepted segment. Higher thresholds filter out small and potentially meaningless regions.	0

computational demands, making it critical to strike a balance between accuracy and efficiency. Given that the total study area encompasses a region of approximately 250 km^2, a representative sub-region of 50 km^2 was selected for controlled testing of hyperparameter effects. Within this test region, different parameter configurations were evaluated using NDVI time series (NDVI-TS) and true/false color composites derived from three distinct PlanetScope (PSB.SD) acquisition dates. The experimental results revealed that the combination of points_per_side = 32 and crop_n_layers = 2 yielded the most accurate and consistent segmentation outcomes. In contrast, parameters such as pred_iou_thresh and stability_score_thresh were observed to have no statistically significant effect on segmentation accuracy, which is consistent with the findings reported by Ferreira et al. (2024). Similarly, other auxiliary parameters like crop_n_points_downscale_factor and min_mask_region_area did not demonstrate meaningful influence on performance and were therefore retained at their default settings. All segmentation experiments were conducted in a cloud-based Google Colab Pro environment equipped with A100 GPU support. Under these conditions, the average processing time per dataset was approximately 6 h, underscoring the computational cost associated with high-resolution, multi-temporal segmentation workflows utilizing complex SAM configurations.

3.9. Post processing

Given that the SAM model operates with an unsupervised segmentation approach, it does not differentiate between thematic classes and thus segments all visually salient objects within the image, including non-agricultural features. As a result, objects such as roads, built-up areas, forest patches, barren lands, and other non-agricultural surfaces are frequently included in the segmentation outputs, introducing a substantial amount of noise and irrelevant segments. This limitation poses a significant challenge for agricultural applications, as directly using the raw segmentation outputs may lead to misinterpretations and reduce the thematic accuracy of the analysis. To address this issue and improve the thematic reliability of the segmentation results, a comprehensive post-processing workflow was implemented. In the first phase of this process, the polygonal features obtained from SAM segmentation were subjected to geometric smoothing using the "Smooth" tool within the QGIS software environment. This operation was applied to remove jagged, pixelated, and unrealistic boundaries, particularly those arising from high-frequency noise in raster-based segmentation. Following the smoothing procedure, a minimum area threshold was defined and polygons smaller than 1000 m^2 were filtered out from the dataset. These small segments were generally found to correspond to non-agricultural artifacts or geometrically insignificant structures. In addition to area filtering, morphological and geometric criteria were employed to further refine the segmentation outputs. Specifically, metrics such as shape regularity, compactness, and convexity were computed for each polygon. Segments that did not satisfy acceptable thresholds for these geometric properties were excluded from the analysis. Through this morphological refinement process, only the polygons exhibiting coherent, compact, and agriculturally relevant geometries were retained. Visually fragmented or irregularly shaped structures, which are less likely to represent actual agricultural parcels, were systematically eliminated (Fig. 10). This post-processing strategy substantially increased the thematic accuracy and practical usability of the SAM-based segmentation results, enabling the extraction of cleaner and more meaningful parcel boundaries that better reflect true agricultural production zones.

4. Results and discussion

4.1. Evaluation metric

In this study, the performance of the segmentation models was comprehensively assessed through a multifaceted evaluation framework, encompassing segmentation capability, classification accuracy, and geometric integrity. To achieve a robust and holistic analysis of model performance, a combination of traditional spatial accuracy metrics and advanced geometric evaluation indicators was employed. In the context of spatial accuracy, four fundamental metrics were utilized: precision, recall, intersection over union (IoU), and their harmonic mean, the F1-score. In this framework, True Positives (TP) refer to the areas where the predicted segmentation and the reference parcels overlap; False Positives (FP) denote the areas segmented by the model that do not correspond to any reference parcel; and False Negatives (FN) represent the regions of reference parcels that were not identified by the model. These metrics collectively provide a quantitative understanding of the model's capability to detect actual agricultural field boundaries in terms of spatial congruence. However, spatial overlap metrics alone may be insufficient for evaluating segmentation quality, particularly in scenarios where structural fragmentation or incorrect merging of adjacent parcels occurs. To address this, two geometric error metrics were incorporated into the analysis: Geometric Over-Segmentation Error (GOSE) and Geometric Under-Segmentation Error (GUSE), as proposed by Persello and Bruzzone (2010). GOSE evaluates the degree to which a single reference object is incorrectly fragmented into multiple predicted segments. This is computed based on the geometric similarity of these fragments to the original object using the IoU metric. GUSE, on the other hand, quantifies the extent to which multiple reference parcels are erroneously merged into a single predicted segment. This measure accounts for the loss of individual parcel identity and evaluates the preservation of structural coherence. Together, these geometric metrics offer a deeper insight into the model's ability to maintain the topological and morphological

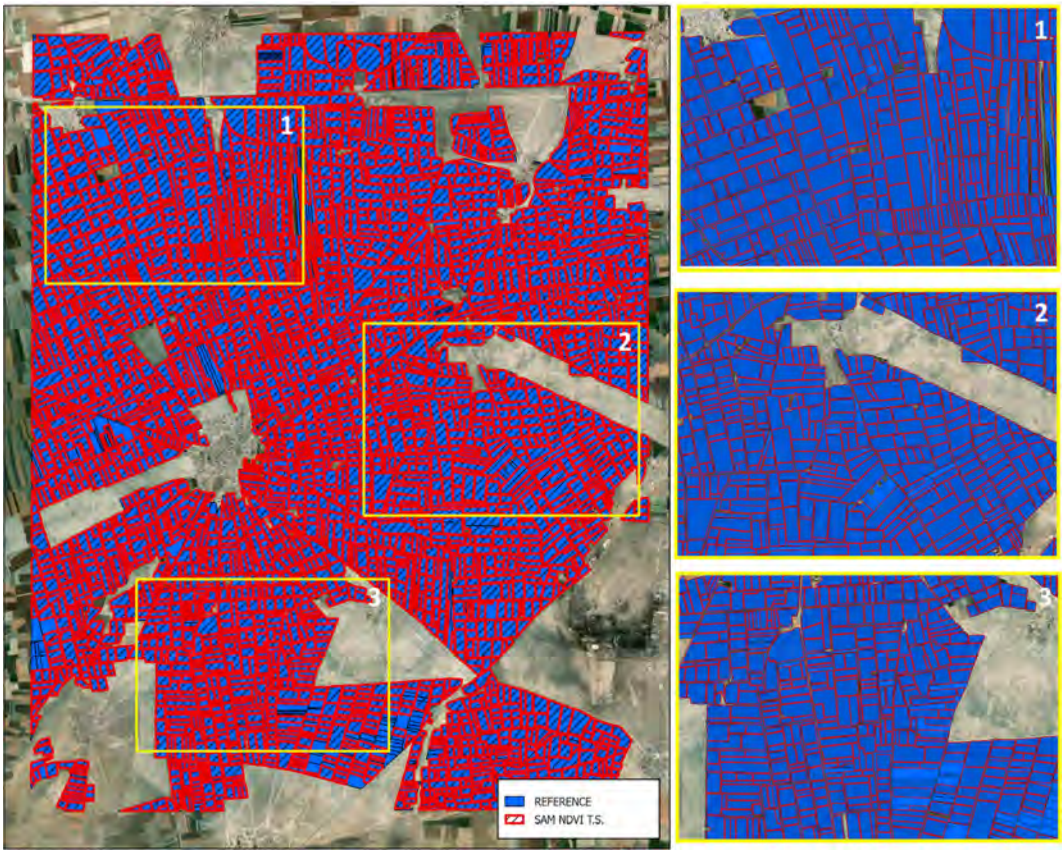

Fig. 10. Overlap of post-processed NDVI time series (NDVI TS) segmentation results with reference parcel boundaries.

integrity of parcel boundaries, thereby complementing traditional accuracy assessments. The mathematical formulations of all performance metrics employed in this study are presented below, while their geometric interpretations and examples are visualized in Fig. 11.

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

$$Recall = \frac{TP}{TP + FP} \quad (2)$$

$$IoU = \frac{TP}{TP + FP + FN} \quad (3)$$

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (4)$$

$$GOSE = \frac{1}{n}\sum\nolimits_{i=0}^{n}\left(1 - \frac{R_i \cap S_i}{R_i}\right) \quad (5)$$

$$GUSE = \frac{1}{n}\sum\nolimits_{i=0}^{n}\left(1 - \frac{R_i \cap S_i}{S_i}\right) \quad (6)$$

TP: True Positive
FP: False Positive
FN: False Negative
n: number of reference segments
$\boldsymbol{S_i}$: denotes the area of the ith reference segment
$\boldsymbol{R_i}$: predicted segment that overlaps the most with R_i.

4.2. Assessment and comparison of SAM results

In this study, it is aimed to detect the boundaries of agricultural parcels with high precision and accuracy using the SAM model. In this direction, firstly, it was aimed to determine the most appropriate date and number of PlanetScope (PSB.SD) images by taking into account the types of crops grown in the study area, the phenological development stages of the crops and the spectral separation characteristics of the crops.

Farmer-declared parcels within the study area boundaries were overlaid with NDVI images of 20 different dates obtained from Sentinel-2 satellite data at 15-day intervals; thus, spectral separation curves specific to each crop type were created. Based on these references, the most appropriate images were selected from among the dates representing different phenological stages for each crop, which could provide highly accurate segmentation on a parcel basis. In this way, the periods with the best spectral separation of the crops and the most distinct crop boundaries due to reflectance-based contrast differences were identified; thus, an ideal input set for the SAM model was created using images from these periods. As part of the segmentation process, pre-processing steps, hyperparameter settings of the model and post-processing methods were applied. The segmentation process was performed using the time series data obtained from the combination of true and false

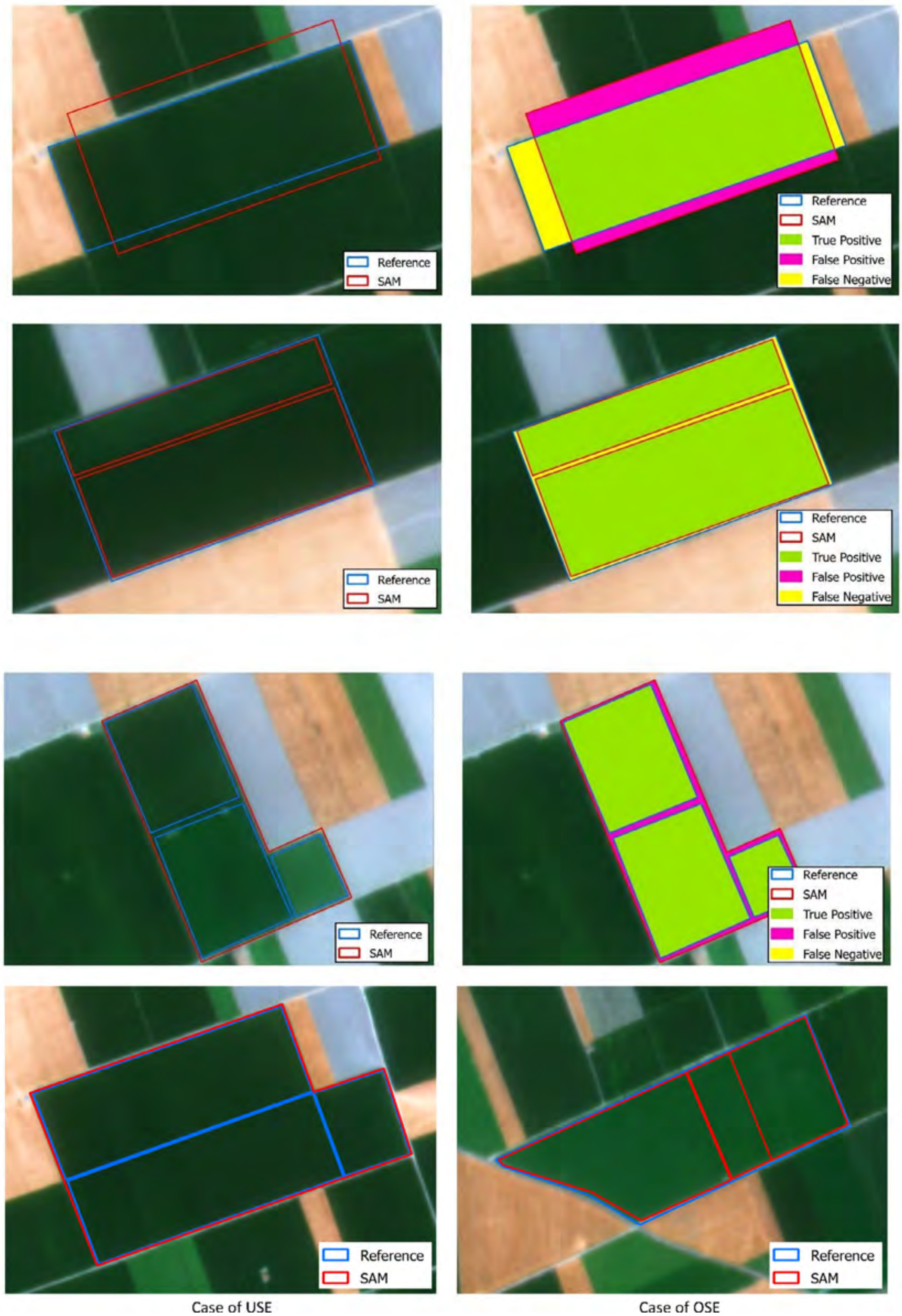

Fig. 11. Spatial representation of TP, FP, FN, GOSE, and GUSE accuracy metrics after the SAM process.

color image combinations of three different dates, as well as the NDVI images generated on these dates. The segmentation results were analyzed both qualitatively and quantitatively in line with the evaluation metrics. Table 4 shows the true color, false color, and NDVI time series values for different dates. Accuracy metrics for segmentation results obtained with color and NDVI time series images are presented. Fig. 12 illustrates the spatial agreement between the segmentation results derived from the NDVI time-series data and the reference parcels, visually presenting the distribution of correctly identified areas as well as the omitted and misclassified agricultural areas.

Table 4
Accuracy evaluation metrics of agricultural parcel segmentation for different data sets.

Data	Precision	Recall	F1	IoU	GOSE	GUSE
23.04.2022 TC	0.97	0.74	0.84	0.73	0.17	0.32
23.04.2022 FC	0.98	0.75	0.84	0.74	0.18	0.32
07.07.2022 TC	0.93	0.85	0.89	0.8	0.25	0.26
07.07.2022 FC	0.94	0.83	0.9	0.8	0.22	0.24
30.09.2022 TC	0.97	0.84	0.9	0.81	0.25	0.27
30.09.2022 FC	0.96	0.85	0.9	0.81	0.22	0.28
NDVI TS	0.98	0.9	0.93	0.89	0.12	0.13

In this study, segmentation performance was analyzed with a multidimensional approach using metrics that reflect both spatial overlap (IoU, Precision, Recall, F1) and geometric/topological integrity (GOSE, GUSE). According to the findings, the NDVI time series (NDVI TS) based multi-temporal approach provided the most successful results in terms of all metrics. The IoU: 0.89, F1: 0.93, GOSE: 0.12 and GUSE: 0.13 values obtained with this method reveal that the segmentation shows high performance in terms of both spatial and geometric accuracy. The low GOSE and GUSE values indicate that the SAM model is able to not only generate the correct number of plots but also to separate them in a topologically consistent manner (Fig. 13).

In the analyses conducted using single-date imagery, the segmentation results for July 7, 2022, and September 30, 2022, were particularly remarkable in terms of spatial accuracy. For these dates, the model achieved high performance metrics, with Intersection over Union (IoU) and F1-scores reaching 0.80 and 0.90, respectively. This success is primarily attributed to the fact that these dates coincide with the peak stages of vegetative development in the region, during which the spectral differences between different crop types are maximized. The increase in spectral separability during these phenologically active periods enabled the model to more clearly delineate parcel boundaries. However, when geometric accuracy is considered, the single-date segmentation results lag significantly behind those obtained using the NDVI time series approach. The calculated GOSE values (0.22–0.25) and GUSE values (0.24–0.28) indicate that the model is prone to both over-segmentation and under-segmentation, compromising the geometric fidelity of the output. The segmentation performance was especially weak for the image dated April 23, 2022. This is due to the phenological conditions of the region: during this period, only winter crops (e.g., wheat, lentils) are actively growing, while most summer crops have not yet emerged. As a result, the spectral signatures of many agricultural parcels remain undefined, and the spatial patterns of crop cultivation are not sufficiently captured in the imagery. In such cases, parcel boundaries are primarily determined by permanent features like in-field roads or parcel edges, which reduces the model's ability to segment fields with high accuracy (Fig. 14). In parcels where no agricultural activity was observed in winter or early spring and where only summer crops were expected to be planted, the SAM model tended to generate block-like segments, often failing to identify the actual internal parcel boundaries. This is reflected in the GUSE value of 0.32 for the April 23 image, indicating substantial under-segmentation. Correspondingly, the segmentation performance for this date was the weakest, with IoU values around 0.73–0.74, GOSE between 0.17 and 0.18, and high GUSE values. These metrics suggest both limited spatial overlap and significant geometric inaccuracies. These findings underscore the inadequacy of relying solely on classical spatial accuracy metrics to evaluate segmentation quality. Metrics such as precision, recall, and IoU may provide useful information on area-based correspondence, but fail to fully capture structural correctness or the preservation of object topology. Therefore, incorporating geometric error metrics such as GOSE and GUSE into the evaluation process enables a more nuanced and comprehensive assessment of model performance. Overall, the results demonstrate that multi-temporal, NDVI-based segmentation approaches offer distinct advantages not only in identifying agricultural areas with greater accuracy but also in generating segmentation outputs with higher geometric consistency and structural integrity.

In the unsupervised segmentation experiments conducted with the SAM model, highly consistent and spatially overlapping results were obtained in agricultural areas characterized by well-defined parcel boundaries and homogeneous internal structures. In such areas, the model's capacity to encode visual cues enabled accurate delineation of boundaries, preserving the geometric integrity of agricultural parcels to a significant extent. However, the segmentation performance deteriorated in fields exhibiting ambiguous boundaries, heterogeneous internal textures, or visually complex spatial patterns. In these cases, the model frequently misclassified some regions as background, resulting in under-segmentation, or conversely, it generated excessive fragmentations through over-segmentation, leading to elevated GOSE and GUSE values. Since SAM relies heavily on visual encoding mechanisms rather than semantic understanding, it is more prone to noise-induced errors in areas where boundary definitions are vague and internal structures lack uniformity. This limitation becomes particularly evident when field boundaries are not demarcated by natural elements, access roads, or neighboring parcels. In such scenarios, SAM tends to erroneously merge multiple distinct parcels into a single segment (i.e., under-segmentation) or split one parcel into numerous small sub-regions (i.e., over-segmentation), thereby reducing the geometric reliability of the output. Furthermore, segmentation accuracy is strongly influenced by the geometric properties of parcels; for instance, fields with elongated or irregular shapes were observed to be more prone to misclassification or incomplete segmentation. Nevertheless, SAM demonstrated strong segmentation performance in clearly delineated, geometrically regular, and actively cultivated parcels. In contrast, uncultivated areas, very small-sized plots, and

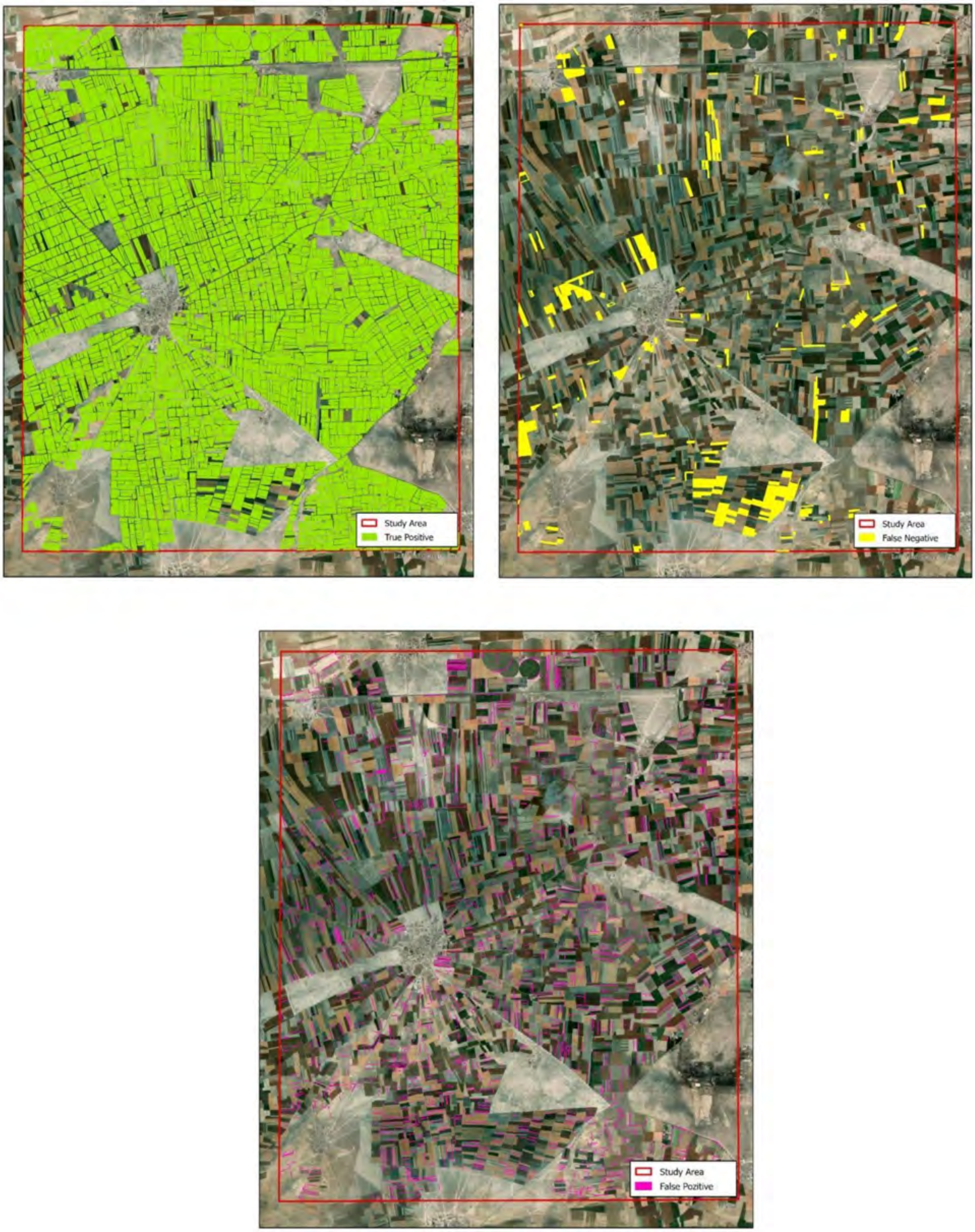

Fig. 12. Comparison of segmentation results derived from NDVI TS data with reference parcels. Shown are true positive (correct), false negative (omission), and false positive (commission) agricultural area.

visually noisy regions were frequently overlooked or incorrectly segmented. This issue is particularly pronounced in the detection of very small parcels, where the spatial resolution of the input satellite imagery becomes a limiting factor. As the minimum mappable unit becomes smaller, higher-resolution imagery is required to achieve precise segmentation results. Ultimately, the success of parcel segmentation using the SAM model is not solely determined

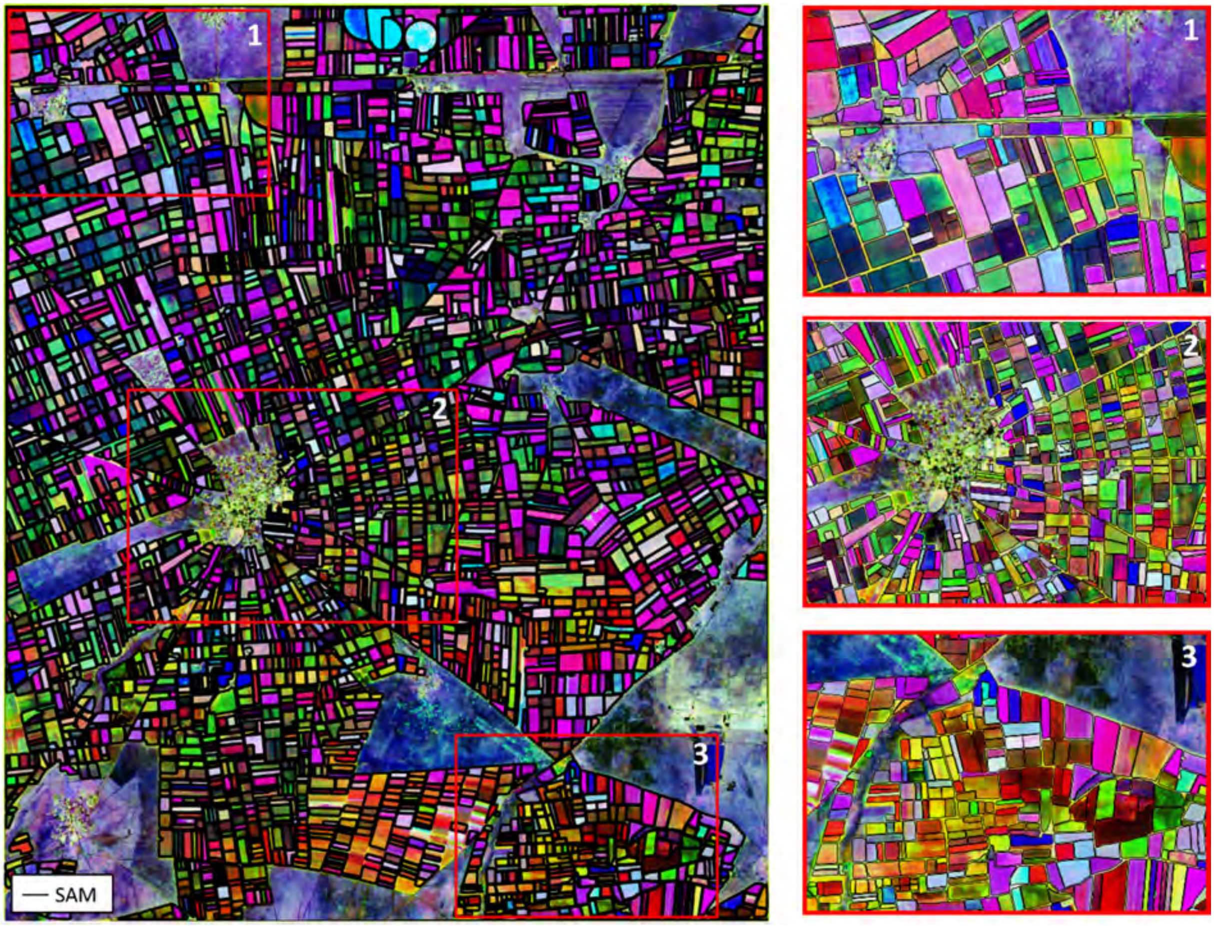

Fig. 13. Agricultural boundary for NDVI TS.

by the model's architecture, but also by several critical external factors: the timing of image acquisition relative to crop phenology, the selection of phenologically representative dates, the use of crop-specific NDVI separation profiles, and the strategic integration of multi-temporal imagery all play essential roles. This study conclusively shows that when such considerations are addressed holistically, the integration of multi-temporal, phenology-aware satellite imagery with state-of-the-art segmentation models like SAM can provide a robust and accurate solution for the automated delineation of agricultural parcel boundaries across heterogeneous landscapes.

5. Conclusion

The accurate and efficient delineation of agricultural land boundaries is of critical importance for the implementation of sustainable agricultural policies, monitoring land use changes, and supporting precision agriculture practices. Accurate boundary extraction not only facilitates the effective monitoring of current agricultural activities but also provides essential baseline data for a wide range of advanced agricultural analyses, including parcel-based production planning, crop diversification studies, phenological stage monitoring, yield estimation, and change detection.However, the heterogeneous nature of agricultural areas, characterized by varying spectral reflectance and complex textural patterns resulting from diverse crop types and phenological stages, poses significant challenges to traditional segmentation approaches. In particular, conventional semantic segmentation methods often fail to localize parcel boundaries accurately due to smooth and ambiguous transitions between adjacent crop types. This limitation renders such methods prone to errors, especially in small and fragmented agricultural systems. In this context, next-generation segmentation approaches, particularly the Segment Anything Model (SAM), offer significant potential for the accurate delineation of cultivated land boundaries. SAM's ability to detect smooth transitions and perform segmentation based on visual continuity enables more accurate and detailed boundary extraction compared to traditional models.

This study highlights that segmentation performance is not solely a function of model architecture, but is also decisively influenced by the content and temporal characteristics of the satellite imagery employed. Understanding the crop calendar and working with images acquired during different phenological stages greatly increases spectral separability among crops, thereby facilitating clearer parcel

Fig. 14. Segmentation results for true and false colors at different times.

boundary delineation. The integration of appropriate vegetation indices, particularly NDVI, enhances surface feature differentiation, while the use of temporal imagery improves segmentation accuracy by more effectively capturing intra-seasonal changes in vegetation dynamics. Empirical findings from the study demonstrate that segmentation based on multi-temporal imagery, particularly NDVI time series (NDVI-TS), yields superior accuracy and consistency compared to single-date analyses. Multi-temporal datasets are particularly effective in addressing spatial–temporal variability, allowing the model to detect subtle changes in field structure and boundary integrity across different crop cycles. SAM's unsupervised and zero-shot capabilities are particularly advantageous in such applications, providing high-fidelity outputs without the need for extensive labeled data.

Nevertheless, the performance of the SAM model is inherently dependent on several key external factors, including the choice of image acquisition date, the spectral and index combinations applied, image preprocessing procedures (e.g., contrast enhancement, edge sharpening), and the tuning of key hyperparameters. Since SAM segments all visually meaningful objects in the image—regardless of semantic relevance—non-agricultural areas such as roads, settlements, or forests may also be included in the segmentation output. To mitigate this, a robust post-processing workflow was implemented, including morphological filtering, minimum area thresholding, and geometric refinement, to isolate only valid agricultural segments. Although the point-based prompt feature of SAM was also tested in this study, its contribution to overall segmentation accuracy was found to be statistically insignificant, consistent with the findings of Huang et al. (2025). Furthermore, as this approach requires manual input, it falls short of achieving fully automatic segmentation. While emerging models such as SAM2 have begun addressing these limitations, further advancements are needed to ensure full adaptability to agricultural production landscapes.The preprocessing and postprocessing strategies developed in this study integrating both spectral and temporal data significantly improved segmentation performance, offering a methodological foundation for future applications involving parcel extraction, change detection, and agricultural monitoring.SAM's strong generalization capacity dramatically reduces the dependence on annotated datasets and is rapidly gaining prominence in remote sensing workflows. Its ability to operate effectively under zero-shot and unsupervised conditions makes it an efficient and low-cost solution for the automatic delineation of cultivated fields, particularly in data-scarce environments. Moreover, the segmentation outputs produced by SAM hold great promise for the acceleration of labeled dataset generation, thereby playing a key role in the training of other deep learning-based models and supporting the data cycle in agricultural remote sensing workflows.

The performance of the SAM model largely depends on the spatial resolution of the input imagery. For small-scale agricultural parcels, the accurate delineation of boundaries requires the use of higher-resolution images. In future studies, the SAM model will be applied to satellite imagery with different spatial resolutions to comprehensively evaluate its segmentation performance across various parcel sizes, as well as the advantages and disadvantages associated with increasing spatial resolution. This analysis aims to establish a methodological framework identifying the optimal image resolution for small-parcel detection in SAM-based segmentation applications. In forthcoming research, the segmentation outputs of the SAM model will also be enhanced through integration with different deep learning approaches, enabling comparative analyses of its performance on satellite imagery with varying spatial resolutions. Furthermore, to assess the model's generalizability in agricultural applications, its performance will be thoroughly investigated in diverse scenarios such as detecting specialized crop patterns (e.g., tea and rice fields), delineating forest stand boundaries, mapping post-fire burned areas, and automatically extracting greenhouse structures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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